



Elastic Collisions in One Dimension

• Worksheet 1

Further Mechanics 1 Chapter 4: Elastic Collisions in One Dimension. Answer all questions. Show your working clearly.
Total: 66 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 4. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ m s}^{-2}$. All particles move on a smooth horizontal surface unless stated otherwise. Use $v_2 - v_1 = e(u_1 - u_2)$ and state your positive direction. Give non-exact answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Newton's law of restitution for two particles	21	
B Collisions with a fixed surface and loss of kinetic energy	20	
C Successive collisions	20	
Roar-level challenge	5	
Total	66	

Section A Newton's law of restitution for two particles

21 marks

1. Two particles A and B are moving in the same direction along a straight line, with speeds 7 m s^{-1} and 3 m s^{-1} respectively, when A collides directly with B . Immediately after the collision, A and B are moving in the same direction with speeds 4 m s^{-1} and 5.5 m s^{-1} respectively. Find the coefficient of restitution between A and B . [2]

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2. A particle A of mass 2 kg moving with speed 6 m s^{-1} collides directly with a particle B of mass 4 kg which is at rest. The coefficient of restitution between A and B is 0.5 . Find the speeds of A and B immediately after the collision. [3]

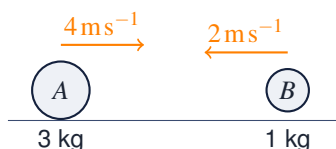
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3. Particles A of mass 3 kg and B of mass 1 kg are moving towards each other along the same straight line, with speeds 4 m s^{-1} and 2 m s^{-1} respectively, when they collide directly. The coefficient of restitution between A and B is 0.5 . [3]



Find the velocities of A and B immediately after the collision.

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4. A particle A of mass m moving with speed u collides directly with a particle B of mass $2m$ which is at rest. The coefficient of restitution between A and B is e . Find, in terms of u and e , the velocities of A and B immediately after the collision, and show that the direction of motion of A is reversed if $e > \frac{1}{2}$. [4]

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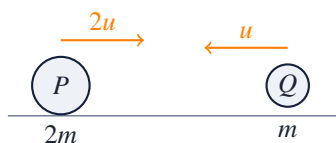
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5. Two particles P and Q , of masses $2m$ and m respectively, are moving towards each other along the same straight line, with speeds $2u$ and u respectively. They collide directly. The coefficient of restitution between P and Q is e .



- (a) Show that the speed of Q immediately after the collision is $u(1 + 2e)$, and find the speed of P immediately after the collision. [4]

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- (b) Given that, immediately after the collision, the speed of Q is four times the speed of P , find the value of e . [3]

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- (c) For this value of e , find the magnitude of the impulse exerted on Q in the collision, in terms of m and u . [2]

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(Total for Question 5 is 9 marks)

**Section B Collisions with a fixed surface and loss of kinetic energy**

20 marks

6. A small ball is dropped from rest from a height of 2.5 m above a horizontal floor. The coefficient of restitution between the ball and the floor is 0.4. [2]
Find the speed with which the ball rebounds from the floor.

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7. A ball of mass 0.2 kg moving horizontally at 10 m s^{-1} hits a fixed smooth vertical wall at right angles, and rebounds with speed 6 m s^{-1} . [3]
Find the coefficient of restitution between the ball and the wall, and the kinetic energy lost in the impact.

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8. A particle A of mass 3 kg moving with speed 5 m s^{-1} collides directly with a particle B of mass 2 kg which is at rest. The particles coalesce. [3]
Find the loss of kinetic energy in the collision.

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9. A particle A of mass 2 kg moving with speed 5 m s^{-1} collides directly with a particle B of mass 3 kg which is at rest. The coefficient of restitution between A and B is 0.6. [4]
Find the speeds of A and B immediately after the collision and the loss of kinetic energy in the collision.

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10. A small ball of mass 0.2 kg is dropped from rest from a height of 2.5 m above a horizontal floor. The coefficient of restitution between the ball and the floor is 0.7 . Air resistance is ignored.

(a) Find the speed of the ball immediately before it first hits the floor. [1]

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(b) Find the height to which the ball rises after its first bounce. [3]

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(c) Find the kinetic energy lost in the first impact. [2]

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(d) Find the time between the first and second impacts with the floor. [2]

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(Total for Question 10 is 8 marks)

**Section C Successive collisions**

20 marks

- 11.** Particles A and B are moving in the same direction along a straight line towards a fixed smooth vertical wall which is perpendicular to the line, with B nearer the wall. A has speed 1.5 m s^{-1} and B has speed 2 m s^{-1} . B hits the wall and rebounds. The coefficient of restitution between B and the wall is 0.5 . [2]

Find the speed of B after it rebounds, and explain why A and B must collide.

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- 12.** Three particles A , B and C , of masses 1 kg , 1 kg and 2 kg respectively, lie at rest in a straight line in that order. A is projected towards B with speed 4 m s^{-1} . The coefficient of restitution between each pair of particles is 0.5 . [3]

Find the speeds of A and B immediately after A collides with B .

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- 13.** For the particles in Question 12, B then collides with C . [3]

Find the speeds of B and C immediately after this collision, and show that there will be a further collision between A and B .

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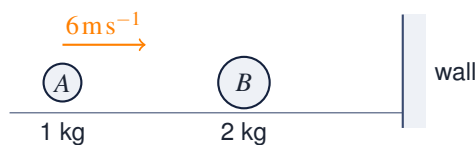
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14. A particle A of mass 1 kg moving with speed 6 ms^{-1} collides directly with a particle B of mass 2 kg which is at rest. The coefficient of restitution between A and B is 0.8 . After the collision B hits a fixed smooth vertical wall perpendicular to its direction of motion. The coefficient of restitution between B and the wall is 0.25 . [4]



Determine whether A and B collide again.

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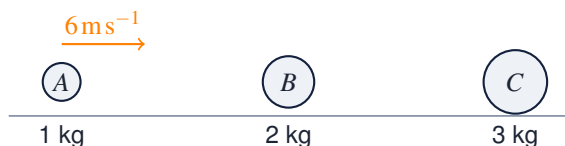
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15. Three particles A , B and C , of masses 1 kg, 2 kg and 3 kg respectively, lie at rest in a straight line on a smooth horizontal surface, in that order. A is projected towards B with speed 6 ms^{-1} . The coefficient of restitution between each pair of particles is $\frac{1}{4}$.



- (a) Find the speeds of A and B immediately after A collides with B . [3]

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- (b) B then collides with C . Find the speeds of B and C immediately after this collision. [3]

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- (c) Show that there will be a further collision between A and B . [2]

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(Total for Question 15 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Three particles A , B and C , of masses m , $2m$ and $3m$ respectively, lie at rest in a straight line on a smooth horizontal surface, in that order. A is projected towards B with speed u . The coefficient of restitution between each pair of particles is e , where $0 < e \leq 1$. [5]
Show that, after B has collided with C , there will be a second collision between A and B if and only if $e < \frac{1}{2}(3 - \sqrt{5})$.

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END OF WORKSHEET

Total: 66 marks